

A Bidomain Model of Epiretinal Stimulation

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Abstract—A computational bidomain model of epiretinal stimulation is presented, consisting of a continuum description of active retinal tissue in contact with bulk vitreous fluid. Results from two-electrode and four-electrode bipolar stimulation suggest that a biphasic cathodic-anodic stimulus sequence is effective in providing targeted focal activation of retinal tissue. Undesired secondary activations beneath each electrode return may be eliminated by using multiple returns for each stimulus electrode.

Index Terms—Bioelectric potentials, charge injection, electrodes, modeling, neuroprosthesis, visual system.

I. INTRODUCTION

DEVELOPMENT of an intraocular vision prosthesis for the blind requires that it be capable of delivering large numbers of parallel electrical stimuli to surviving retinal neurons in order to facilitate the generation of useful visual percepts. Such stimulation can be achieved via a regular array of bipolar stimulus current sources comprised of multiple current injection and return electrodes, not necessarily equal in number. The ensuing complex stimulus regimes, however, may lead to unsatisfactory dispersion of retinal activation or even the unintended generation of multiple visual percepts [1]. In this study, we present a bidomain mathematical model of epiretinal bipolar stimulation, which allows for the prediction of patterns of retinal activation arising from multiple stimulus and return electrodes.

Existing models of epiretinal stimulation are not suited to examine patterns of retinal activation in bulk volume of active retinal tissue, since they are comprised only of segments of single retinal ganglion cells (RGCs) and axon fibers, each coupled resistively with neighboring neural segments [2]–[4]. In such models, extracellular voltages induced by the stimulus electrodes are time-independent, determined only by the electrode boundary conditions, since transmembrane currents are assumed to not influence the extracellular potential [5]. The presence of a bulk volume of active retinal tissue near the stimulating electrodes suggests that some extracellular current would be diverted into depolarizing the cell membranes, such that the extracellular voltage would not be fixed in time, even if a constant stimulus current was applied. This was experimentally confirmed in 1956 by Brindley [6], who observed

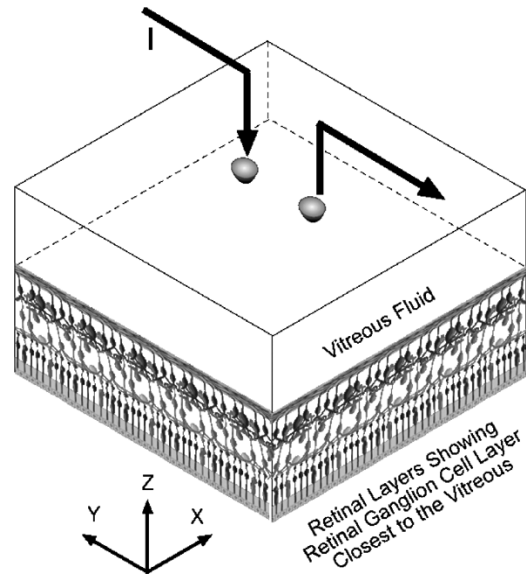


Fig. 1. Schematic diagram of bidomain retinal model showing a single pair of hemispherical electrodes located epiretinally, injecting current I into the vitreous fluid and retinal neurons. A layer of active retinal tissue consists of RGCs coupled intracellularly in the Z -direction only. Dimensions of the overall rectangular domain are $4 \times 4 \times 0.6$ mm (not drawn to scale). Other key dimensions include: retinal layer thickness—0.2 mm; electrode radius—0.2 mm; distance from lower electrode points to upper retinal surface—0.2 mm; distance between electrode centers—1 mm. For the two-electrode configuration shown, electrodes are placed along the mid-plane perpendicular to X and symmetrically located about the mid-plane perpendicular to Y .

characteristic rounding of voltage pulses during square-wave constant-current stimulation of active retinal tissue.

To investigate retinal activation patterns resulting from bipolar current stimulation, we have formulated a simplified bidomain model of bulk retinal tissue, epiretinally stimulated by one or two pairs of electrodes providing biphasic currents (Fig. 1). Bidomain models provide a continuum description of coupled intracellular and extracellular spaces within excitable tissue, and have already proven useful in cardiac electrophysiological modeling [7]–[9]. From bidomain theory, extracellular currents are able to stimulate excitable cells only if the anisotropic ratios of intracellular and extracellular conductivities are unequal [10]. Although the anisotropic conductivity ratios for retinal tissue have not yet been experimentally determined, it can be argued that such an unequal anisotropy ratio in both domains is evident in that RGC and bipolar cells are intracellularly coupled preferentially in the vertical direction normal to the retinal surface (Fig. 1), with minimal synaptic coupling in the lateral directions. If substantial lateral coupling were present, retinal tissue would form a functional syncytium much like cardiac tissue, and contrary to observation, RGC action potentials would propagate laterally. Note that in Fig. 1, we have not included an uppermost layer of axons overlaying the active RGC layer, since epiretinal stimulation does not

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appear to activate axons from peripheral sites [1], [11]. Thus, we utilize a simplified representation of active retinal tissue, consisting of RGCs coupled intracellularly in one direction.

II. METHODS

A. Model Formulation

To investigate the activation patterns of retinal tissue resulting from epiretinal current injection, simulations were undertaken using a three-dimensional (3-D) computational model comprised of bulk vitreous fluid and an active bidomain retinal volume consisting of coupled intracellular and extracellular domains (Fig. 1). Hemispherical electrodes were placed in the vitreous fluid a small distance above the retinal surface. In the present study, the following electrode configurations were examined: a single pair consisting of one injection and one return electrode, as well as a four electrode configuration arranged symmetrically in the corners of a square. In this latter configuration, both two injection and two return, as well as one injection and three return electrodes were simulated. In all simulations, the centers of the return electrodes were connected to ground (zero volts).

Within the epiretinal vitreous region, the Poisson governing equation for the extracellular voltage distribution V_e was

$$-\nabla \cdot (\sigma_b \nabla V_e) = I \quad (1)$$

where σ_b is the conductivity of the bulk vitreous medium and I is the volume current density injected into the vitreous at the given location [12]. The value of σ_b was taken to be 1.28 S/m, equivalent to the resistivity of $78 \Omega \cdot \text{cm}$ measured for physiological saline [13]. Using hemispherical electrodes and constant σ_b , (1) expands to the following system of equations within the vitreous domain:

$$\nabla^2 V_e = \frac{-3i_s^n}{2\sigma_b \pi r^3} \quad \text{within the } n\text{th injecting electrode} \quad (2)$$

$$\nabla^2 V_e = \frac{3i_r^m}{2\sigma_b \pi r^3} \quad \text{within the } m\text{th return electrode} \quad (3)$$

$$\nabla^2 V_e = 0 \quad \text{outside all electrodes} \quad (4)$$

with essential boundary condition

$$V_e = 0 \quad \text{at all return electrode centers} \quad (5)$$

where i_s^n , i_r^m are the current injected and returned (in amperes) from the n th injecting and m th return electrodes respectively, and r is the electrode radius. In general, the i_s^n currents injected are given, whereas the return currents i_r^m are unknown when there is more than one return electrode. These return currents can be determined from the additional constraint given by (5).

Within the active retinal region, the bidomain equations [7], [9] linking the extracellular and intracellular voltages were formulated as

$$\sigma_e \nabla^2 V_e = -I_m \quad (6)$$

and

$$\sigma_{i,z} \frac{\partial^2 V_i}{\partial z^2} = I_m \quad (7)$$

TABLE I
MODEL PARAMETERS

V_m	-70 mV	c	0.0097
$[\text{Ca}]_i$	0.1 μM	n	0.0577
m	0.0096	a	0.0248
h	0.9644	h_A	0.5843

TABLE II
INITIAL VALUES

Symbol	Description	Value	Units
σ_b	Conductivity of vitreous fluid	1.28	S/m
σ_e	Extracellular retinal conductivity	0.02	S/m
$\sigma_{i,z}$	Intracellular retinal conductivity in z-direction	0.02	S/m
β	Surface to volume ratio	166,670	m^{-1}
C_m	Membrane capacitance	1	$\mu\text{F}/\text{cm}^2$
g_{Na}	Membrane conductance of i_{Na}	70	mS/cm^2
g_{Ca}	Membrane conductance of i_{Ca}	1.5	mS/cm^2
g_{K}	Membrane conductance of i_{K}	18	mS/cm^2
g_{KCa}	Membrane conductance of i_{KCa}	54	mS/cm^2
g_{L}	Membrane conductance of i_{L}	0.065	mS/cm^2
g_{L}	Membrane conductance of i_{L}	0.0295	mS/cm^2
V_{Na}	Na reversal potential	50	mV
V_{K}	K reversal potential	-77	mV
V_{L}	Leakage reversal potential	54.3	mV
$[\text{Ca}]_{\text{res}}$	Resting cytosolic Ca concentration	0.1	μM
$[\text{Ca}]_e$	Extracellular Ca concentration	2	mM
τ_{Ca}	Ca removal time constant	1.5	ms
λ_r	Ca buffer volume-to-surface ratio	600	nL/cm^2
F	Faraday constant	96.48	C/mmol
R	Universal gas constant	8.31	$\text{J}/\text{mol}^\circ\text{K}$
T	Absolute temperature	310	K

where σ_e is the extracellular conductivity within the retinal layer, $\sigma_{i,z}$ is the intracellular conductivity in the z direction alone (intracellular conductivities in the x , y directions are taken to be zero), and I_m is the RGC membrane current density per unit volume, given by

$$I_m = \beta \left(C_m \frac{\partial V_m}{\partial t} + J_{\text{ion}} \right) \quad (8)$$

where β is the surface to volume ratio, C_m and J_{ion} are the membrane capacitance and ionic current per unit area, respectively, and V_m is the transmembrane potential given by

$$V_m = V_i - V_e. \quad (9)$$

Values for retinal σ_e and $\sigma_{i,z}$ were 0.02 S/m, equivalent to the resistivity of $5000 \Omega \cdot \text{cm}$ measured in frog retina [13]. Note that these conductivities are considerably lower than the retinal conductivities used in previous models [2], [4], [14], which simply assumed retinal conductivities similar to that of physiological saline. Assuming a cylindrical retinal neural cell geometry with a diameter of $24 \mu\text{m}$, the surface-to-volume parameter β is determined as $1.6667 \times 10^5 \text{ m}^{-1}$. Values for all model parameters can be found in Table I.

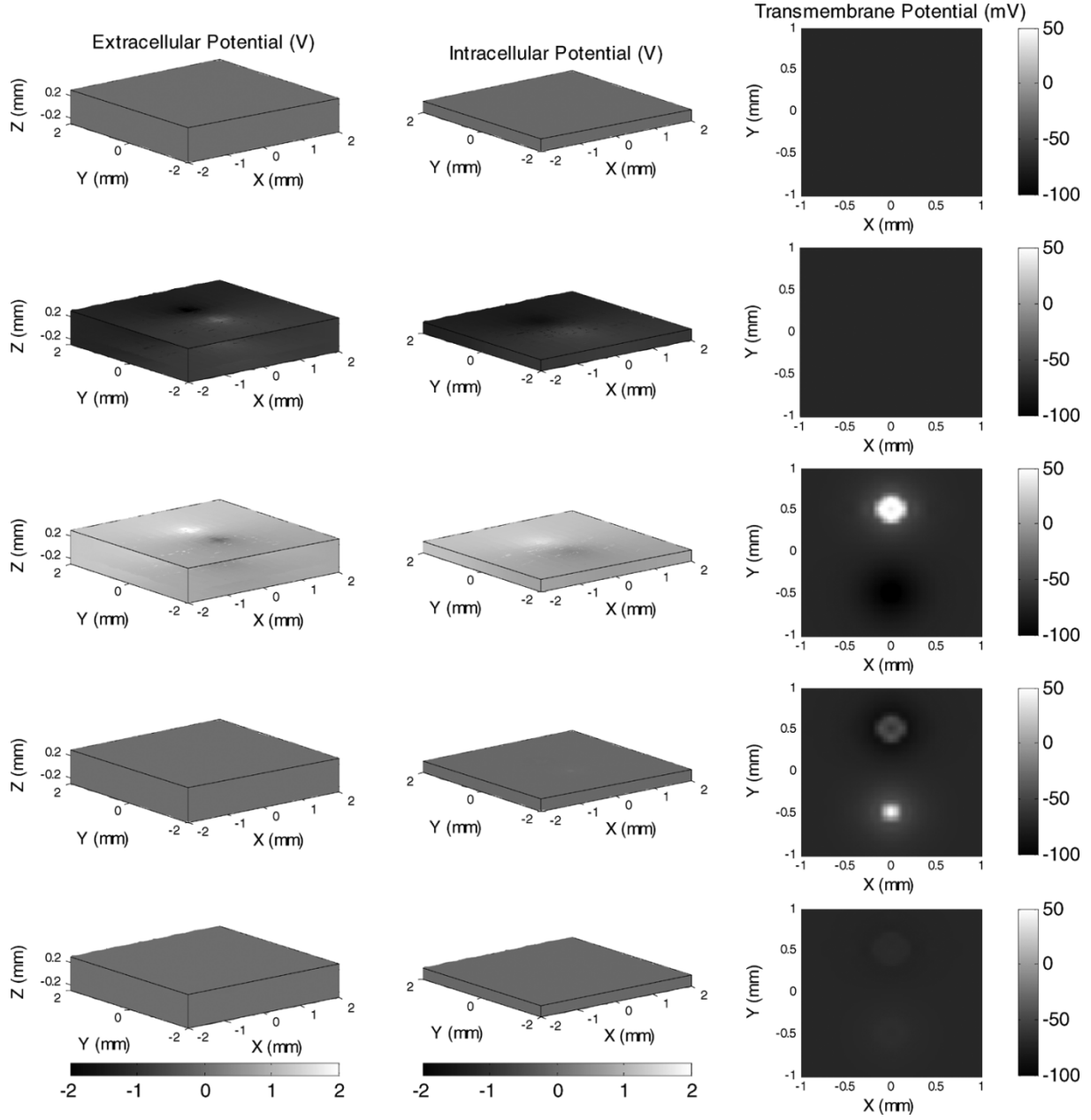


Fig. 2. Results of biphasic cathodic-anodic two-electrode epiretinal stimulation. A constant-current of 1-mA magnitude was applied at $t = 2$ ms, with the injecting electrode located at $x = 0$ mm, $y = +0.5$ mm. Initial stimulus current at the injecting electrode was such that this electrode was cathodic relative to the return electrode. Current polarity was reversed at $t = 3$ ms, with stimulation terminated at $t = 4$ ms. The center of the return electrode was always held at 0 V. From top to bottom, each row illustrates time snapshots at $t = 1$ (top), 2, 3, 4, and 5 ms (bottom) of the extracellular voltage distribution (left), the intracellular voltage distribution in the retinal layer (middle), and a zoomed-in view of the transmembrane potential in the mid-retinal layer located 100 μ m below and parallel to the retinal surface (right). For the 3-D volume plots, the z axis has been magnified for clarity. Grayscale bars below the 3-D volume plots are in units of volts, while grayscale bars to the right of transmembrane potential plots are in units of millivolts.

To determine J_{ion} in (8), we used a modified RGC model based on the formulation of Fohlmeister and Miller [15]

$$\begin{aligned}
 J_{\text{ion}} = & \overbrace{g_{Na} m^3 h (V_m - V_{Na})}^{i_{Na}} + \overbrace{g_{Ca} c^3 (V_m - V_{Ca})}^{i_{Ca}} \\
 & + \overbrace{g_K n^4 (V_m - V_K)}^{i_K} + \overbrace{g_A a^3 h_A (V_m - V_K)}^{i_{K_a}} \\
 & + \overbrace{g_{K,Ca} (V_m - V_K)}^{i_{K,Ca}} + \overbrace{g_L (V_m - V_L)}^{i_L} \quad (10)
 \end{aligned}$$

where m , h , c , n , a , and h_A are gating variables governed by first-order voltage dependant kinetics according to

$$\begin{aligned}
 \frac{dx}{dt} &= \alpha_x (1 - x) - \beta_x x \\
 x &\equiv m, h, c, n, a, h_A \quad (11)
 \end{aligned}$$

with rate constants α_x , β_x being functions of transmembrane potential V_m , and given in Fohlmeister and Miller [15]. All other parameters in (10) are fixed and given in Table I, with the exception of V_{Ca} , the equilibrium potential for i_{Ca} , which

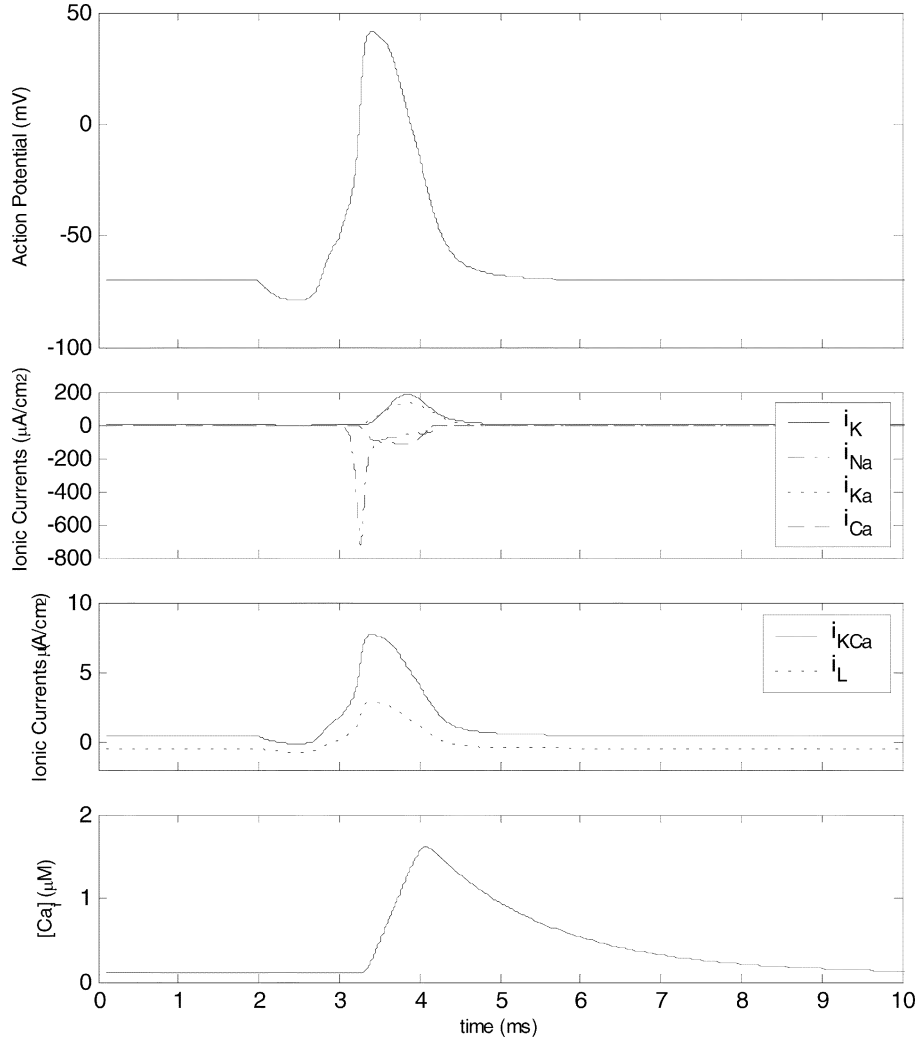


Fig. 3. Action potential activated by extracellular biphasic stimulus (top panel), underlying membrane currents (middle panels), and cytosolic calcium concentration (lower panel) of RGC located in the mid-retinal layer below the injecting electrode in the simulation of Fig. 2.

is determined from the cytosolic calcium concentration $[Ca]_i$, according to the Nernst equation

$$V_{Ca} = \frac{RT}{2F} \ln \left(\frac{[Ca]_e}{[Ca]_i} \right) \quad (12)$$

where the extracellular calcium concentration $[Ca]_e$ is fixed at 2 mM. $[Ca]_i$ is determined from first-order buffering kinetics according to

$$\frac{d[Ca]_i}{dt} = \frac{-3I_{Ca}}{2F\lambda_r} - \frac{([Ca]_i - [Ca]_{res})}{\tau_{Ca}}. \quad (13)$$

Finally, in order to solve (2)–(9), the following zero-flux boundary conditions were enforced on V_e and V_i :

$$\frac{\partial V_e}{\partial x} = 0, \frac{\partial V_e}{\partial y} = 0, \frac{\partial V_e}{\partial z} = 0 \quad (14)$$

were imposed on the x , y , and z edges of the domain, respectively

$$\frac{\partial V_i}{\partial x} = 0, \frac{\partial V_i}{\partial y} = 0, \frac{\partial V_i}{\partial z} = 0 \quad (15)$$

were imposed on the x , y , and z edges of the smaller retinal domain, respectively, and an additional flux-continuity boundary

condition was applied at the retinal/vitreous interface on the upper retinal surface due to the discontinuity in extracellular conductivity from vitreous to retina

$$\underbrace{\sigma_e \frac{\partial V_e}{\partial z}}_{\text{retinal region}} = \underbrace{\sigma_b \frac{\partial V_e}{\partial z}}_{\text{vitreous bulk region}}. \quad (16)$$

Table II lists initial values used for all RGC variables for every cell in the retinal domain.

B. Numerical Implementation

The system of (1)–(16) was solved in the time-domain using a custom second-order variable-step predictor-corrector algorithm, adopting the stable method of Victorri *et al.* [16] for all gating variable updates. At each time step, a leap-frog scheme was utilized with spatial differentiation performed using a custom spectral collocation method. In brief, the 3-D domain was segmented into a number of smaller box regions enclosing each hemi-spherical electrode as well as the spaces between electrodes and between electrodes and the domain boundary. Within each box region, a nonuniform x - y - z mesh was generated using Chebyshev node spacing [17], [18]. This resulted in a more dense packing of nodes near each segment boundary

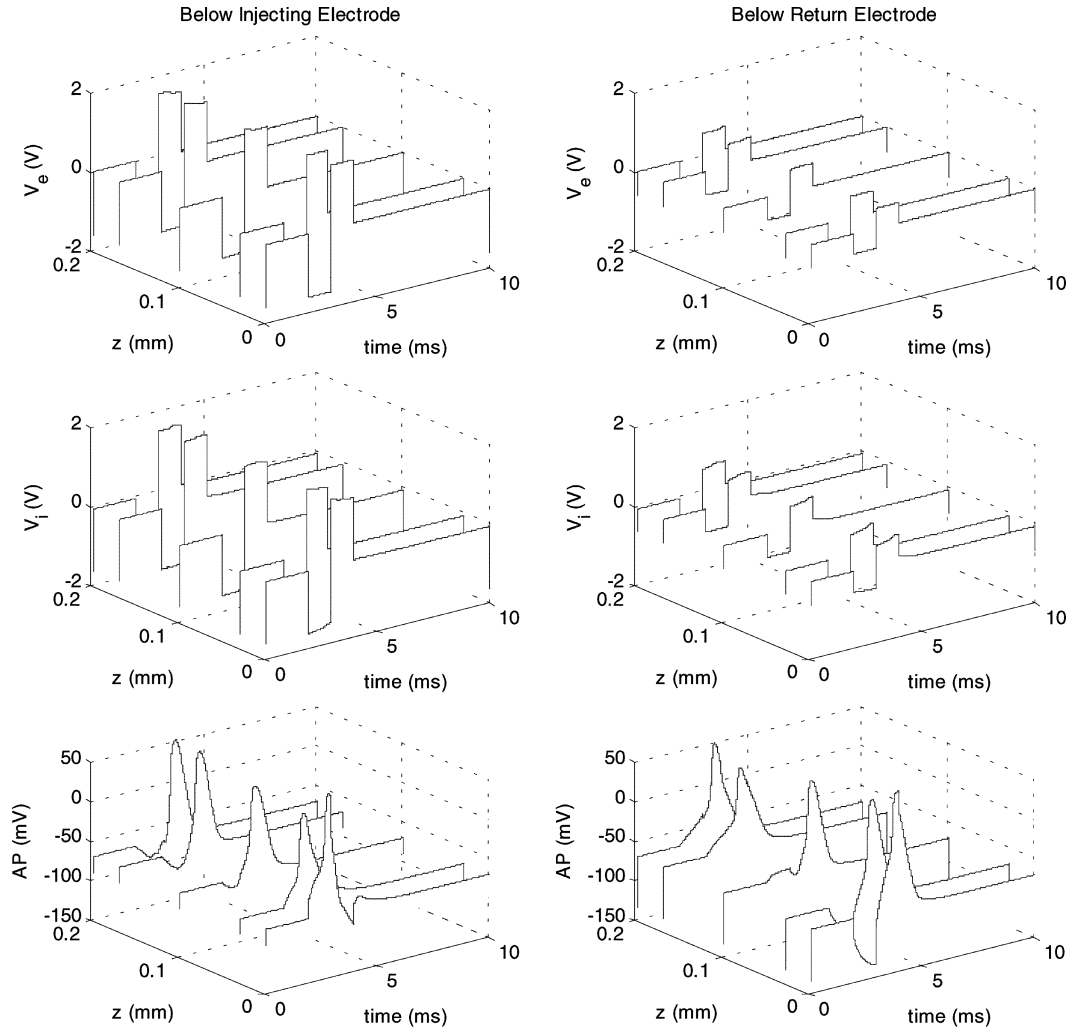


Fig. 4. Representative time-varying waveforms for the simulation shown in Fig. 2. Extracellular voltages (top row), intracellular voltages (middle row), and action potential waveforms (lower row) throughout the retinal domain directly beneath the centers of the injecting (left column) and return (right column) electrodes. The z axis in units of millimeters represents the distance below the upper retinal surface nearest the electrodes.

compared with the centers of each segment. First-order derivative continuity in extracellular and intracellular voltage was enforced at each segment boundary. The resulting system of differentiation matrices generated was then inverted to solve for the voltage at each node. Spectral differentiation allowed for more accurate derivatives with fewer nodes than the more traditional finite-difference approach [18]. For the simulations presented in this study, five nodes were placed in each x and y direction within the electrodes, and 11 nodes in the inter-electrode spaces and between the outermost electrodes and the boundary of the domain. In the z direction, due to its smaller size, only three nodes were placed within electrodes and five nodes each between the electrode and upper retinal surface and within the retinal layer itself. In addition, five extra nodes were added in z to represent the intracellular retinal domain. That is, a total of $25 \times 39 \times 16$ nodes in x , y , and z , respectively, for the two-electrode simulations, and $39 \times 39 \times 16$ nodes for the four-electrode simulations. This modest number of nodes allowed the models to be solved within a few hours on a 2.66-GHz Pentium IV (Intel Corporation, Santa Clara, CA) desktop workstation.

III. RESULTS

A. Two-Electrode Epiretinal Stimulation

Fig. 2 illustrates 1-ms time snapshots of the extracellular potential, intracellular retinal potential, and retinal activation patterns arising from 1-mA biphasic bipolar epiretinal stimulation. The stimulus sequence was cathodic-anodic relative to the grounded return electrode. Fig. 3 illustrates the associated action potential (AP) and ion currents generated for a representative RGC located vertically below the center of the current injecting electrode in the mid-retinal layer, while Fig. 4 illustrates the time-varying extracellular, intracellular, and transmembrane voltage waveforms directly below both electrodes. From these results, we see that a focal region of activation is generated directly below the cathodic injecting electrode during this first phase of stimulus delivery. Cells nearest the injecting electrode are rapidly depolarized on stimulus onset at $t = 2$ ms (Fig. 4, lower left), due to the cathodic stimulus. That is, a negative step in extracellular potential will tend to depolarize the cell membrane, as can be seen from (9). The no-flux intracellular

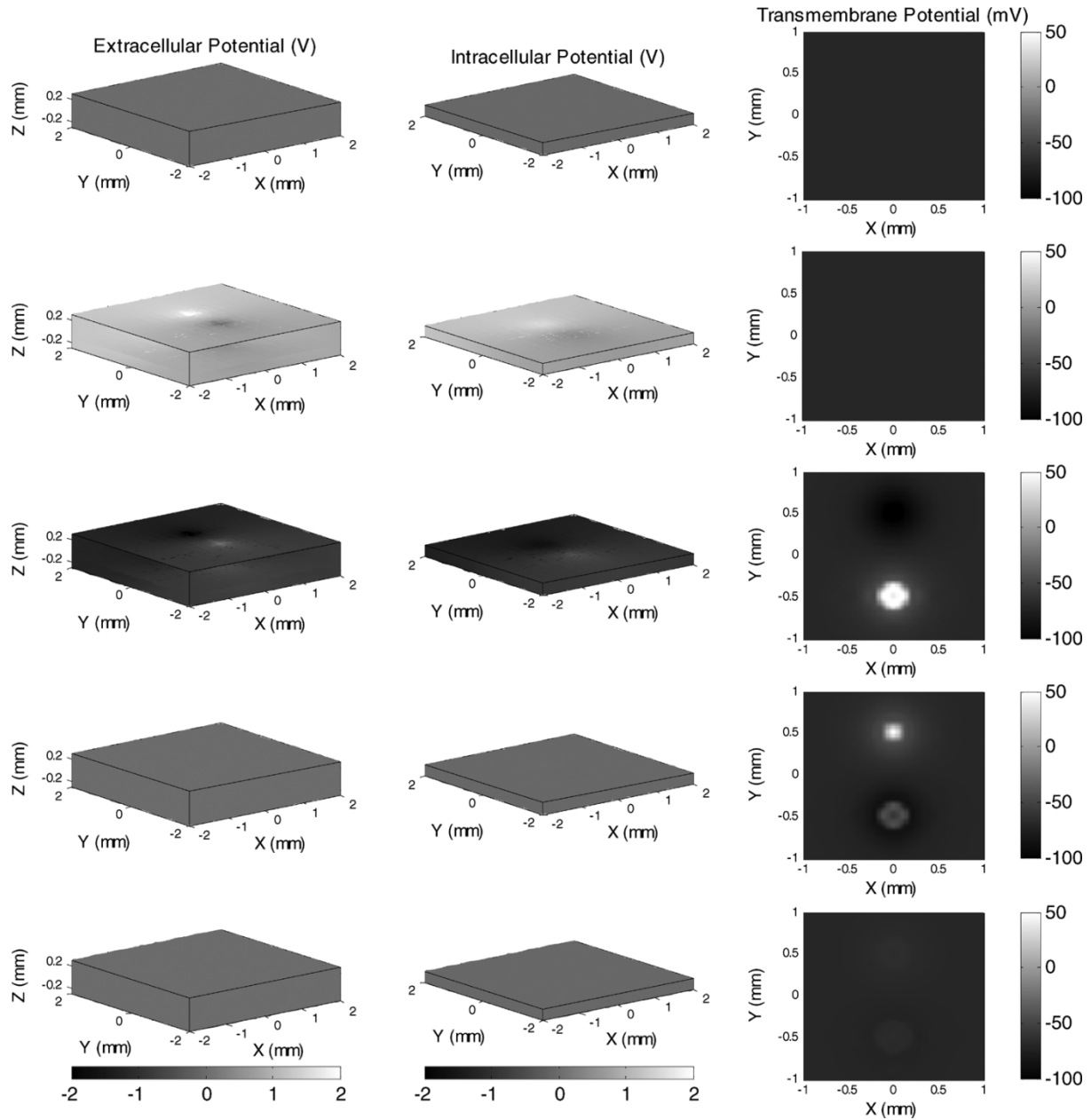


Fig. 5. Simulation results of anodic-cathodic two-electrode epiretinal stimulation. All model parameters and figure panel layouts are identical to that of Fig. 2, with the exception that the biphasic stimulus polarity has now been reversed.

boundary conditions, (15), imply that total membrane current is conserved throughout the retinal domain. Thus, the initial depolarizing current from the cathodic stimulus is offset by a passive hyperpolarization in cells further below the retinal surface (Fig. 4, lower left). Even these cells however, elicit an AP due to excitation of the RGCs nearest the retinal surface. For the return electrode however, the opposite is seen to hold. The first phase of stimulus delivery leads to hyperpolarization of cells nearest the surface and depolarization of deeper cells. On stimulus reversal, there is a rapid depolarization of cells nearest the retinal surface to produce an undesired secondary focal spot of activation beneath the return electrode (Figs. 2 and 4).

The simulation results of Fig. 5 illustrate the same two-electrode configuration, but with the stimulus polarity reversed to an

anodic-cathodic sequence. In this case, the cells beneath the return electrode are activated during the first stimulus phase, with a secondary focal spot of activation beneath the injecting electrode on stimulus reversal.

In order to attempt to eliminate the undesired secondary focal activation on stimulus reversal shown in Figs. 2 and 5, another simulation was performed using a monophasic cathodal stimulus. Fig. 6. illustrates the results applied to the same two-electrode configuration used previously. As expected, a focal spot of activation is still produced immediately below the injecting electrode. However, the secondary focal activation is still present (below the return electrode) on stimulus offset. In this case, however, the depolarizing current is produced by the discharge in cell membrane capacitance across the ionic channels in (8).

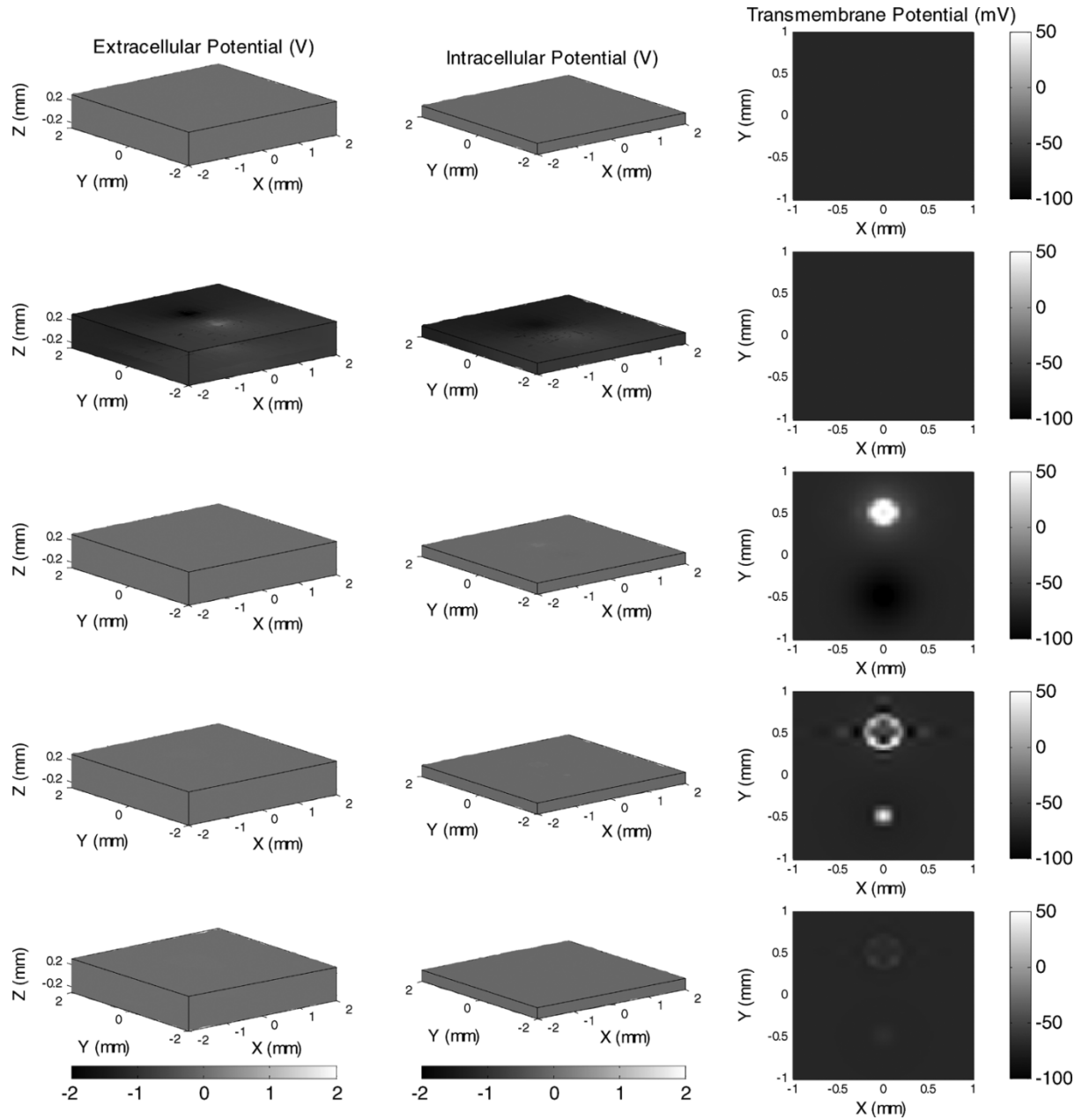


Fig. 6. Simulation results of cathodic monophasic two-electrode epiretinal stimulation. A monophasic constant-current waveform of 1 mA magnitude was applied at $t = 2$ ms, with the injecting electrode located at $x = 0$ and $y = +0.5$ mm. The stimulus at the injecting electrode was cathodic relative to the return electrode. Stimulation was terminated at $t = 3$ ms. All other model parameters and figure panel layouts are identical to that of Fig. 2.

B. Four-Electrode Epiretinal Stimulation

In order to examine the pattern of retinal activation produced by multiple electrode pairs, two simulations were undertaken using four electrodes arranged symmetrically on the corners of a 1-mm square. Fig. 7 illustrates the results of parallel application of 1-mA biphasic cathodic-anodic stimulation delivered to two diagonally opposing injecting electrodes, with the other two return electrodes grounded. As before, focal spots of activation are produced immediately below the injecting electrodes. In addition, the secondary activations below both returns are still present. To eliminate these, a final simulation was carried out using the same four-electrode configuration, but this time using only one injecting electrode and three returns. Results are shown

in Fig. 8. As can be seen, the focal activation is produced below the injecting electrode. This time, however, the return current is dispersed over three electrodes, resulting in diminished charging of local cell membrane capacitances beneath the returns. As a result, the secondary activation below the returns on stimulus polarity reversal has now been eliminated.

IV. DISCUSSION AND CONCLUSION

The bidomain model of active retinal tissue presented in this study is able to generate patterns of electrical activation within bulk retinal tissue when subjected to epiretinal biphasic stimulation. As expected, extracellular voltage distribution exhibited time-varying behavior during each phase of constant current

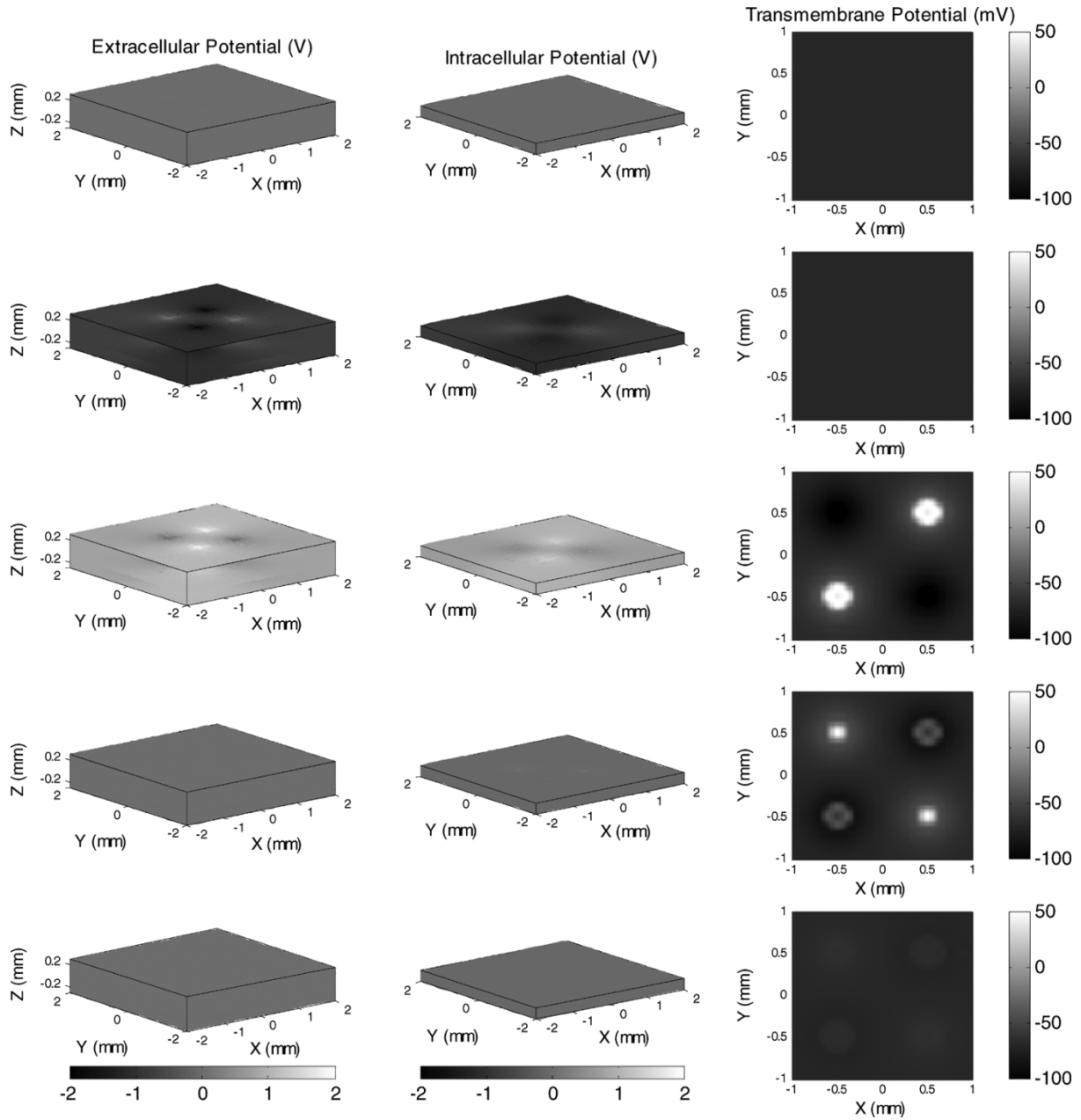


Fig. 7. Simulation results of cathodic four-electrode epiretinal stimulation. Biphasic constant-currents of 1 mA were applied simultaneously at $t = 2$ ms, with injecting electrodes located at $x = +0.5, y = +0.5$, and $x = -0.5, y = -0.5$ mm. The remaining two return electrodes were held at 0 V throughout the simulation. Initially on stimulus delivery, the two injecting electrodes were cathodic relative to the returns. Current polarity in the injecting electrodes was reversed at $t = 3$ ms, and reset to zero at $t = 4$ ms. All figure panel layouts are identical to Fig. 2.

stimulation, due to the transmembrane current flow within the active retinal tissue (Fig. 4). Simulation results for two- and four-electrode configurations suggest that a cathodic-anodic biphasic sequence of epiretinal stimulation is effective in producing directed focal spots of activation below each stimulus (injecting) electrode, as opposed to anodic-first stimulation, in broad agreement with existing data on epiretinal stimulation [19], [20]. Undesired secondary activations were present during polarity reversal beneath each bipolar return electrode. Interestingly, multiple visual percepts have also been reported from blind patients during epiretinal bipolar stimulation through a single pair of electrodes, although the mechanism for this was unclear [1]. Results of this study suggest that parallel bipolar

epiretinal stimulation is effective in providing targeted retinal activation, without secondary activations, provided only that multiple returns are used for each stimulus electrode. This indicates that epiretinal bipolar electrode arrangements will need to incorporate this feature into their design. The presence of such secondary activations suggests that monopolar stimulation using a distant return electrode may be preferable, since the results of this study show that only focal areas under the stimulus electrodes are activated. However, disadvantages of monopolar stimulation include the fact that it requires an additional distinct return electrode with its associated manufacturing and/or implantation complexities [21]. Also, depending on this electrode location, the return path may be such that

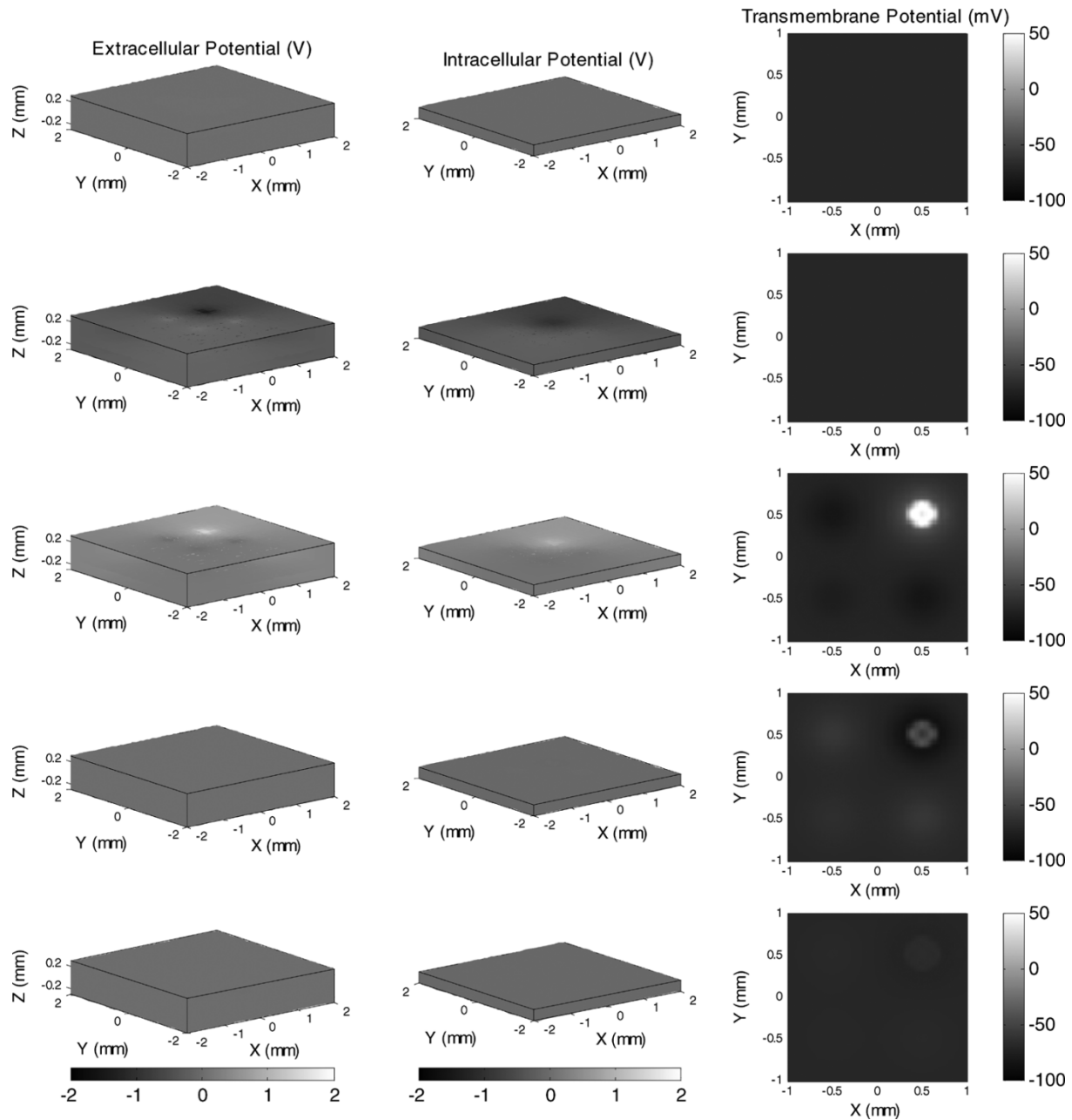


Fig. 8. Results of biphasic cathodic-anodic four-electrode epiretinal stimulation with one stimulus and three returns. A biphasic current of 1-mA magnitude was applied at $t = 2$ ms into the injecting electrode located at $x = +0.5$ and $y = +0.5$ mm. The remaining three return electrodes were held at 0 V throughout the simulation. Initially on stimulus delivery, the injecting electrode was cathodic relative to the three returns. Current polarity in the injecting electrode was reversed at $t = 3$ ms, and reset to zero at $t = 4$ ms. All figure panel layouts are identical to Fig. 2.

current flow away from the retinal surface directly to the return is favored, requiring higher stimulus currents. Alternatively, bipolar arrays are easier to manufacture, with tighter control being maintained over the stimulation since fewer variables are involved.

A possible limitation of the present study is the assumption of a highly idealized continuum description of bulk retinal active tissue consisting of neural cells with zero lateral conductivity. This assumption ignores the presence of known lateral pathways involving amacrine and horizontal cells [22], which play an important role in parallel processing within the retina during normal vision [23]. Epiretinal electrical stimulation, however,

has been reported to produce distinct focal phosphenes in human patients [11], suggesting that lateral conductivity is likely to be minimal under these conditions. As to how appropriate the bidomain model is for characterizing bulk retinal tissue, this still needs to be experimentally verified. Nonetheless, preliminary data suggesting the presence of multiple visual percepts during bipolar stimulation [1], as well as the observation of voltage pulse rounding during constant current stimulation of bulk retinal tissue [6], are consistent with the results of this study. We anticipate that the continuum retinal model presented here will be used to investigate the efficacy of even more complex electrode arrangements as part of a future visual prosthesis design [24].

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